



UM Study System

A week-aware companion to Spectrum that ends the daily file-drowning.

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Spectrum is the University of Malaya's academic material portal. It works, but it hides what you actually need behind long, undifferentiated file lists. This document is the working spec for a small web app that fixes that by filtering files by week, with a one-click sign-in and a deliberately small UI.

1. The problem

Every day a UM student opens Spectrum, signs in, and is met with a long, alphabetised list of **course subjects + course codes**. That list itself is already heavier than it needs to be — there are courses on it that the student isn't actively studying that week, and codes that look almost identical to one another. The cognitive cost of just *finding the right course* is non-zero.

Once a course is opened, the situation gets worse. Spectrum dumps **every file the lecturer has ever uploaded** into one stream: project briefs, Week 1 slides, Week 2 readings, sample code, tutorial sheets, past-year papers, ad-hoc announcements. By Week 9 or Week 12 the list is so long that the student has to scroll through everything older just to reach this week's lecture slides.

What happens in practice: the student opens the page, sees the wall of files, decides they'll do it later, and switches to YouTube or Discord. The friction is small per visit, but it compounds across a 14-week semester and ends up being the reason material gets read the night before the exam.

Symptoms, in plain terms

- Daily overwhelm at the course-list screen.
- Information overload after entering a course — everything shown, nothing prioritised.
- Scrolling cost grows linearly with the week; by Week 10+ it's significant.
- Loss of attention → loss of momentum → procrastination.

2. What I want to build

A small web app that sits on top of Spectrum. The student logs in once with their UM Google account, picks a course, picks a week, and sees **only the files for that week**. Nothing else. No project folder unless they ask for the project folder. No past weeks unless they ask for a past week.

Everything in the app is built around two ideas: **filter by week**, and **show as little as possible at a time**. The UI follows Claude's visual language — warm neutral background, a single coral accent, generous whitespace, no decorative chrome.

Core promise

From login to the right file in three taps.

Sign in → pick course → pick week → done.

Why simplicity is non-negotiable here

- The whole point is to **reduce** what the student sees, not add another dashboard.
- Every extra feature is a chance for the user to stall and switch apps.
- Claude-style UI: lots of white space, one accent colour, no badges/popups/banners.
- If a feature can't be reached in one tap from the current screen, it doesn't ship in v1.

3. Features

Feature	What it does
One-click Google sign-in	Sign in with @um.edu.my Google account. No password screen. Other domains are rejected.
Course picker	Grid of the student's enrolled courses. Each card shows the course code, course name, and course image.
Week picker	Weeks 1–14 as a flat grid. Current week is highlighted and reachable via a single 'Jump to week' button.
Filtered file list	Once a week is chosen, only files whose name or section tag matches that week are displayed.
Manual section override (v1)	Alongside the 14 week buttons, the student can pick named sections like Project, Assignment, etc.
Autonomous section scanning (v2)	Backend scans Spectrum sections for a given course and learns which section belongs to which week.
File-type filter chips	Inside a week's file list, optional chips narrow further by type (slides / notes / recordings / etc.).
Direct download	Each file has one button: Download. No previews, no quick-look, no extra UI.

4. Login: one-click @um.edu.my

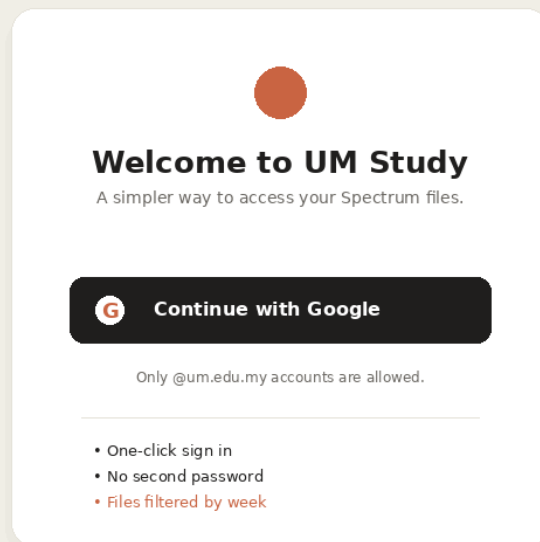
Sign-in is the only thing in the app that could feel like friction, so it has to be as small as possible. The whole login screen is one card and one button.

How it works

- Frontend triggers a Google OAuth 2.0 flow (single button, 'Continue with Google').
- On the OAuth callback, the backend inspects the returned **email** claim.
- If the email does not end in **@um.edu.my**, the session is rejected with a single message.
- If it does, the user record is created (or fetched) and a session cookie is set.
- No second password, no email verification step, no extra form.

Implementation sketch

```
# pseudo-code, server-side callback
email = google_token.email
if not email.endswith("@um.edu.my"):
    return 403, "Use your UM Google account."
user = User.get_or_create(email=email)
session.set(user_id=user.id)
```



University of Malaya — Internal student tool

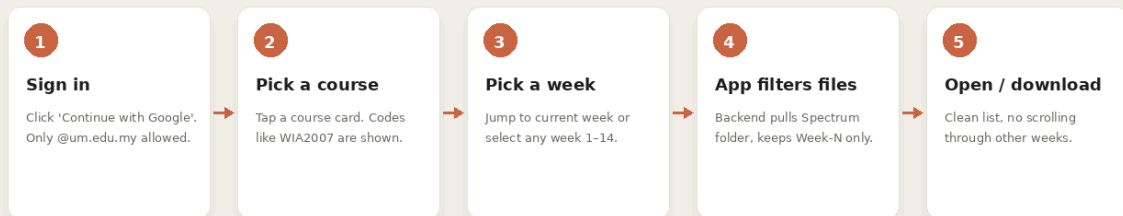
Mockup — login screen. One card, one button, one rule.

5. User flow

The full path from launching the app to looking at this week's files is five steps, and four of them are a single tap.

User flow

From sign-in to the filtered file list — five steps, nothing extra.



Manual section override (e.g. 'Project') is one tap away on the week-picker screen.

User flow — five steps, with Steps 2 and 3 doing the bulk of the work.

Step-by-step

- **1. Sign in.** Google OAuth, @um.edu.my only, one button.
- **2. Pick a course.** Cards showing course code + name. Tap one.
- **3. Pick a week.** 'Jump to current week' or any of weeks 1–14. Other named sections (Project, Past Year Papers, custom) live in the same screen as small pills.
- **4. App filters files.** Backend pulls the course's Spectrum sections and keeps only the files tagged for the chosen week.
- **5. Open / download.** Clean file list, with optional type chips (Slides / Notes / Recordings / Other). One Download button per row.

6. UI mockups

Four screens, end to end. Visual language follows Claude: cream background, coral accent, dark ink text, a single font family, plenty of breathing room.



Your courses

Semester 2, 2025/2026

WIA2007

Software Modeling

Dr. Lim

Open →

WIA2003

Data Structures

Dr. Tan

Open →

WIB1002

Mathematics II

Dr. Wong

Open →

WIA2002

Database Systems

Dr. Rahman

Open →

GIG1003

Critical Thinking

Dr. Aisha

Open →

WIA2005

Operating Systems

Dr. Kumar

Open →

Screen 2 — course picker. Grid of course cards, no notification badges or counts.

[← Back to courses](#)

Pick a week

We'll only show files tagged with that week.

Jump to current week — Week 9

Week 1 past 4 files	Week 2 past 5 files	Week 3 past 6 files	Week 4 past 7 files	Week 5 past 8 files	Week 6 past 4 files	Week 7 past 5 files
Week 8 past 6 files	Week 9 current 8 files	Week 10 upcoming	Week 11 upcoming	Week 12 upcoming	Week 13 upcoming	Week 14 upcoming

Other sections

Project

Assignments

Past Year Papers

Custom...

Screen 3 — week picker. 'Jump to current week' is the primary action; 'Project', 'Assignments', 'Past Year Papers' and a 'Custom' pill cover the non-week sections.

← Pick a different week

Week 9 materials

Showing 8 files tagged Week 9. Everything else is hidden.

All Slides Notes Recordings Other

PDF	Week09_Lecture_Slides.pdf Slides · 2.4 MB · uploaded Apr 14	Download ↓
PDF	W9_Notes_UseCase.pdf Notes · 1.1 MB · uploaded Apr 14	Download ↓
DOC	Week9_Tutorial.docx Tutorial · 320 KB · uploaded Apr 15	Download ↓
MP4	Recording_Week9_Lecture1.mp4 Recording · 412 MB · uploaded Apr 14	Download ↓
PDF	W9_Reading_Booch.pdf Reading · 3.8 MB · uploaded Apr 12	Download ↓
ZIP	Week9_CodeSamples.zip Code · 740 KB · uploaded Apr 14	Download ↓
PDF	W9_Quiz_Brief.pdf Quiz · 210 KB · uploaded Apr 16	Download ↓
PDF	Week09_Summary.pdf Summary · 640 KB · uploaded Apr 17	Download ↓

Screen 4 — filtered file list. Only files tagged with the chosen week are shown. Optional chips narrow by type. One Download button per row, nothing else.

7. How sections are handled

Spectrum organises course material into sections. The naming is uneven: some courses use 'Week 1', 'Week 2', ... some use 'Lecture Week 1' or 'W1', and most courses also have non-week sections like 'Project' or 'Past Year Papers'. The plan is to handle this in two phases.

Phase 1 — manual mapping (v1, ships first)

- Predefined section names: **Week 1 ... Week 14, Project, Assignments, Past Year Papers**.
- For courses that use non-standard section names, I add a custom mapping per course: e.g. 'Lecture Week 1' → Week 1, 'Tutorial W3' → Week 3.
- Mappings live in a simple JSON file per course (or one DB table), edited by me at first and later by the student themselves through a small settings screen.
- Files inside a mapped section inherit the week tag automatically.

Phase 2 — autonomous section scanning (v2)

- Backend job walks every section of every enrolled course on a schedule.
- Each section title is run through a small classifier: detects week numbers (Week 1, W1, Lecture 1, etc.) and named buckets (project, past year, assignment).
- Confidence threshold: high-confidence matches are auto-applied, low-confidence matches surface a one-tap confirmation in the UI.

- File names themselves are also scanned (e.g. *W9_Notes_UseCase.pdf* → tagged Week 9) as a fallback when the section name is ambiguous.
- End state: the student never has to type or pick a section mapping.

8. Design principles

The aesthetic is Claude's — but more important than the look is the discipline around what gets shown. Below are the rules I'll use to push back on feature creep.

Show less, by default.	Anything that doesn't help the student act on this week's material is hidden.
One primary action per screen.	Sign in. Pick course. Pick week. Download file. Each screen has one obvious next step.
Use one accent colour.	Coral (#C96442) for primary actions and the current week. No green-yellow-red badges.
Whitespace before chrome.	Before adding a divider or a panel, see if more space achieves the same thing.
No notifications.	No red dots, no 'NEW' badges, no toast pop-ups. The list of files is the notification.
Cap the list.	If the filtered list is longer than ~15 items, that's a signal the filter is too coarse.

Colour tokens

	Background	#F0EEE6
	Card	#FFFFFF
	Accent	#C96442
	Ink	#1F1E1D
	Muted text	#6B6863
	Line	#E5E2D9

9. Build plan

Roughly in the order I'll build it. Each milestone is small enough to ship on its own.

#	Milestone	What's in it
M1	Auth shell	Google OAuth set up, @um.edu.my domain check, session cookie, empty home s
M2	Spectrum integration	Server-side helper that, given a student's session, can list enrolled courses and the
M3	Course picker UI	Render the course list as cards. Claude-style design tokens applied.
M4	Manual section mapping + week picker	JSON mapping file. Week picker grid with current-week jump. 'Project
M5	Filtered file list	Only files for the chosen week are shown. Type chips. One Download button per fi
M6	Polish + caching	Cache Spectrum responses so the app feels instant. Empty states. Error toasts.
M7	Autonomous section scanning	Background job + classifier. Promote to default once confidence is high enough.

Out of scope (for now)

- Chat / messaging / forum features.
- Calendar sync, deadline tracking, grade pulling.
- Mobile-native app (the web app should be responsive; that's enough).
- Sharing files between students — the source of truth is still Spectrum.

End of document. Next action: M1 (auth shell) — wire Google OAuth and the @um.edu.my domain check, then deploy a single empty page behind it.